

HBSM: Hybrid Boundary Sensitivity Mechanisms

Anonymous authors

Paper under double-blind review

Abstract

This is a living paper shell for a preregistered Mac-first experiment. HBSM tests whether sensitivity in frontier hybrid reasoners can be explained and predicted cheaply. The broad forward-only sensitivity claim is wounded by recent related work, so no novelty or positive result is claimed yet.

1 Current Gate

The branch must pass sensitivity heterogeneity, no-forward predictor, and mechanism gates. If it overlaps with HORN, it should be folded into a unified hybrid-quantization paper.

2 Mac Artifact Status

A deterministic synthetic B1/B2 packet validates the layer-sensitivity and cheap predictor readout. It reports kurtosis-versus-sensitivity Spearman rho 0.657 and two boundary-flagged top-decile hits, producing `SYNTHETIC_PASS_REAL_LAYER_SENSITIVITY_NEXT`. This is synthetic-only: it is not model evidence, not a novelty claim, and not GPU evidence.

3 Next Evidence

The next admissible row is a real layer-sensitivity sweep on current hybrid reasoners. If cheap predictors fail to correlate with forward sensitivity, HBSM should be killed or folded into HORN rather than kept as a separate paper.